Do Increases in Minimum Wage Affect Unemployment?

A panel data analysis of U.S. state-level minimum wage and unemployment rates, 1984–2020

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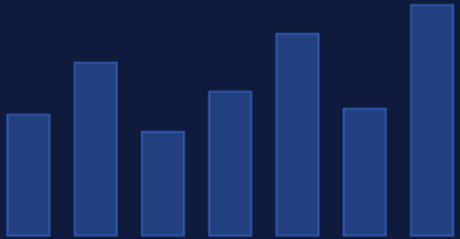

Why this Project

- The effect of minimum wage on unemployment is a hot debate
- It was an easy panel problem since each state sets their own wage floors at different times
- Personal stake – I have a job

Research Question

Why does this debate matter?

↑

Labor Costs Rise

Firms face higher costs per worker as minimum wages increase. Standard theory predicts reduced labor demand.

≈

Mixed Evidence

Card & Krueger (1994) found no disemployment effects; Neumark & Wascher (2007) find negative effects across many studies.

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The Question

Do increases in the real effective minimum wage lead to measurable increases in state-level unemployment rates?

Data & Variables

Panel dataset: 50 U.S. states × 37 years (1984–2020), n = 1,850 observations

Variable	Description	Source
Unemployment Rate	Annual avg. % unemployed by state	U.S. Census
Real Min. Wage	Effective min. wage (2020 dollars)	Kaggle Dataset
Median HH Income	State median household income	FRED
log(Income)	Log of median HH income	Derived

1,850
State-Year Observations

50
U.S. States

37
Years (1984–2020)

5.62%
Avg Unemployment Rate

Descriptive Statistics

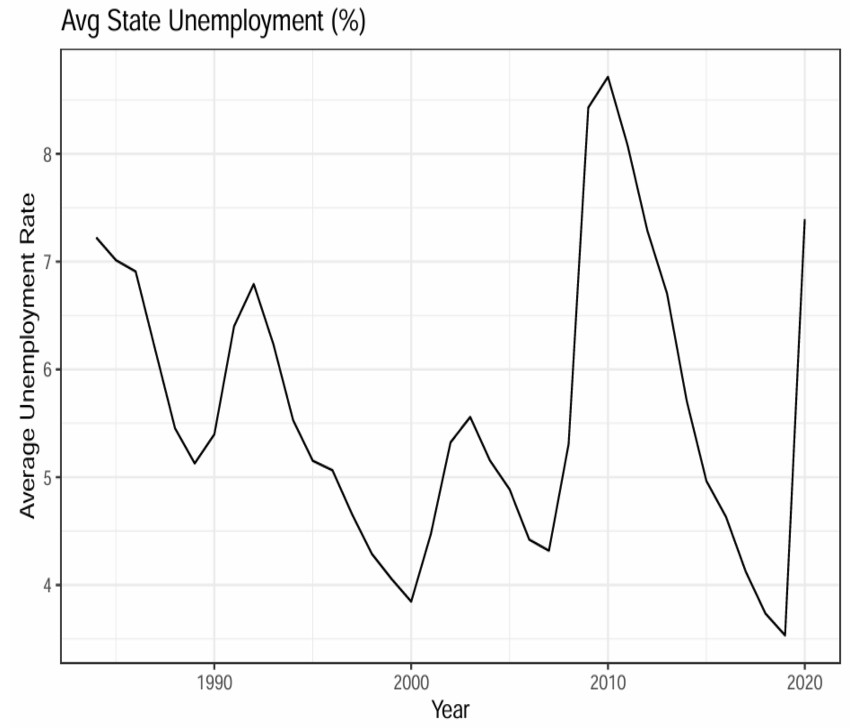
Summary of key variables across all 1,850 state-year observations

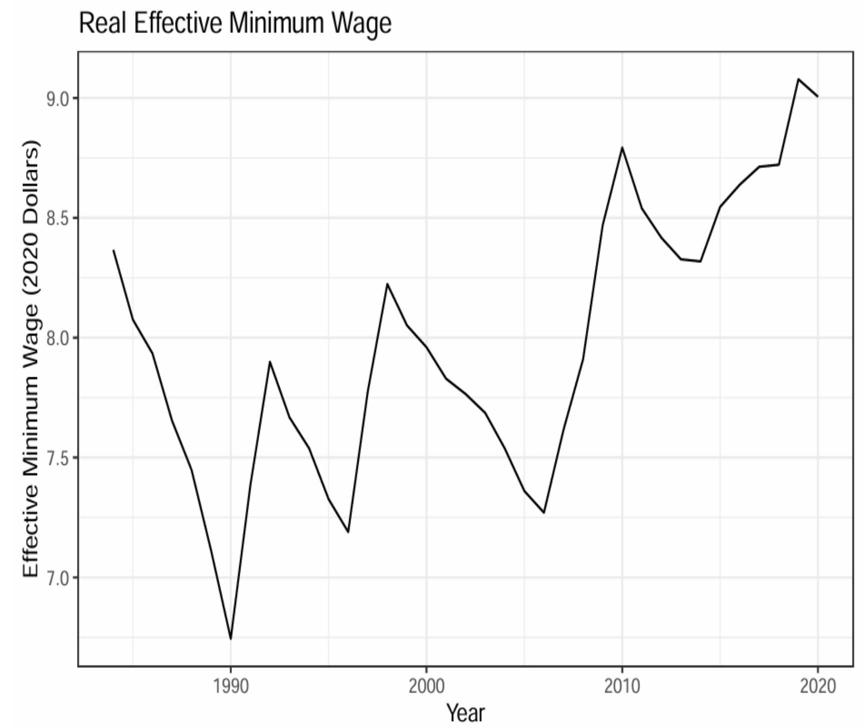
	Min	1st Quarter	Median	3rd Quarter	Max
Unemployment Rate (%)	2.10%	4.25%	5.29%	6.70%	14.08%
Real Effective Min. Wage ($2020)	$6.42	$7.32	$7.84	$8.34	$13.66
Log of Median Household Income ($)	10.62	11.01	11.13	11.25	11.66

- Substantial variation in unemployment across states and time supports panel methods.
- Real minimum wage ranges from $6.42 to $13.66, reflecting policy and inflation differences.
- Unemployment peaks are visible during the early-1990s recession and the 2008 Great Recession.

Trends Over Time

Average unemployment and real minimum wage, 1984–2020

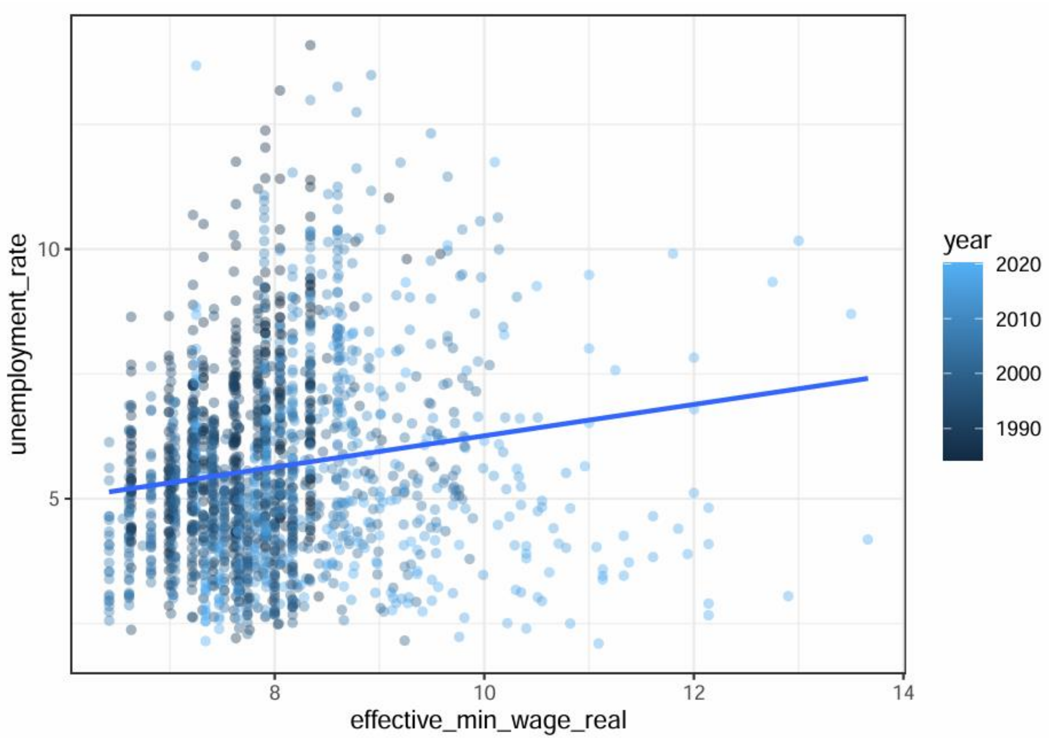




Note: Unemployment spikes align with the early-1990s recession and 2008 Great Recession. Minimum wage follows a more gradual policy-driven trend.

Correlation

Average unemployment and real minimum wage per state, 1984–2020



Empirical Strategy

Three models building from simple OLS to two-way fixed effects

1

Pooled OLS

$$\text{Unemployment}_{it} = \beta_0 + \beta_1\ \text{MinWage}_{it} + \varepsilon_{it}$$

Baseline model. Captures raw correlation but ignores state-level differences and time trends.

2

OLS + Income Control

$$\text{Unemployment}_{it} = \beta_0 + \beta_1\ \text{MinWage}_{it} + \beta_2\ \text{Income}_{it} + \varepsilon_{it}$$

Adds median household income to control for variation in economic conditions across states.

3

Two-Way Fixed Effects

$$\text{Unemployment}_{it} = \beta_1\ \text{MinWage}_{it} + \beta_2\ \text{Income}_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

Preferred model. Adds state FE (α_i) and year FE (λ_t) to control for unobserved differences within states and shared time shocks.

Regression Results

Effect of real effective minimum wage on state unemployment rate

Min. Wage Coeff.	0.314	0.660	0.159
(Std. Error)	(0.047)	(0.047)	(0.039)
Income Coeff.	—	-6.68e-5	-5.84e-5
State FE	No	No	Yes
Year FE	No	No	Yes
R^2	0.024	0.171	0.765
N	1,850	1,850	1,850

0.314

Model 1 — raw effect

Without controls

0.660

Model 2 — rises with income

Cross-section bias revealed

0.159

Model 3 — within-state effect

Best Model

All p-values are below .001

Limitations & Future Directions

What this study can and cannot tell us

≠ Correlation, Not Causation

Fixed effects reduce omitted variable bias but do not fully identify causal effects. Minimum wage changes may be correlated with unobserved time-varying economic conditions such as cost of living and population changes.

↔ Reverse Causality

States may raise minimum wages when their economies are strong (or weak), creating reverse causality. Without an instrument or natural experiment, this cannot be resolved.

Σ Aggregation Bias

State-level unemployment combines across different labor markets. Effects on teens or low-skilled workers may differ substantially from effects on the overall rate.

→ Future Research

Apply difference-in-differences with border state pairs or regression discontinuity at wage thresholds for stronger causal identification.

Conclusion

- Higher minimum wages are positively associated with unemployment across all models.
- The preferred model suggests a 1-dollar real wage increase is linked to ~0.16 pp higher unemployment.
- Results should be interpreted as conditional associations, not causal effects.

Key Finding

0.159

pp increase in unemployment per $1 real wage increase (within-state, FE model)